

Information Processing (8 th std)

Choose the Appropriate Answer

1. In a class there are 26 boys and 15 girls. The teacher wants to select a boy or a girl to represent a quiz competition. In how many ways can the teacher make this selection?

- (A) 41 (B) 26 (C) 15 (D) 390

Ans: (A) 41

2. How many outcomes can you get when you toss three coins once?

- (A) 6 (B) 8 (C) 3 (D) 2

Ans: (B) 8

3. In how many ways can you answer 3 multiple choice questions, with the choices A, B, C and D?

- (A) 4 (B) 3 (C) 12 (D) 64

Ans: (d) 64

4. How many 2 digit numbers contain the number 7?

- (A) 10 (B) 18 (C) 19 (D) 20

Ans: (b) 18

5. What is the eleventh Fibonacci number?

- (a) 55 (b) 77 (c) 89 (d) 144

Ans: (c) 89

F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	F_{11}	F_{12}	F_{13}	F_{14}	F_{15}	F_{16}	F_{17}	F_{18}
1	1	2	3	5	8	13	21	34	55	89	144	233	377	610	987	1597	2584

6. If $F(n)$ is a Fibonacci number and $n = 8$, which of the following is true?

- (a) $F(8) = F(9) + F(10)$ (b) $F(8) = F(7) + F(6)$
(c) $F(8) = F(10) \times F(9)$ (d) $F(8) = F(7) - F(6)$

Ans: (b) $F(8) = F(7) + F(6)$

7. Every 3rd number of the Fibonacci sequence is a multiple of _____

- (a) 2 (b) 3 (c) 5 (d) 8

Ans: (a) 2

8. Every _____ number of the Fibonacci sequence is a multiple of 8

- (a) 2th (b) 4th (c) 6th (d) 8th

Ans: (c) 6th

9. The difference between the 18th and 17th Fibonacci number is

- (a) 233 (b) 377 (c) 610 (d) 987

Ans: (d) 987

10. Common prime factors of 30 and 250 are
(a) 2×5 (b) 3×5 (c) $2 \times 3 \times 5$ (d) 5×5
Ans: (a) 2×5

Prime factors of 30 are $2 \times 3 \times 5$
Prime factors of 250 are $5 \times 5 \times 5 \times 2$
Common prime factors are 2×5

11. Common prime factors of 36, 60 and 72 are
(a) 2×2 (b) 2×3 (c) 3×3 (d) $3 \times 2 \times 2$
Ans: (d) $3 \times 2 \times 2$

Prime factors of 36 are $2 \times 2 \times 3 \times 3$
Prime factors of 60 are $2 \times 2 \times 3 \times 5$
Prime factors of 72 are $2 \times 2 \times 2 \times 3 \times 3$
Common prime factors are $2 \times 2 \times 3$

12. Two numbers are said to be co-prime numbers if their HCF is
(a) 2 (b) 3 (c) 0 (d) 1
Ans: (d) 1

13. A group of letters are given. A numerical code has been given to each letter.
These letters have to be unscrambled into a meaningful word. Find out the code
for the word so formed from the 4 answers given.

L I N C P E
1 2 3 4 5 6

- (a) 2 3 4 1 5 6 (b) 5 6 3 4 2 1 (c) 6 1 3 5 2 4 (d) 4 2 1 3 5 6
Ans: (b) 5 6 3 4 2 1
14. If the word 'PHONE' is coded as 'SKRQH', how will 'RADIO' be coded?
(a) SCGNH (b) VRNGG (c) UDGLR (d) SDHKQ
Ans: (c) UDGLR

15. Online or television advertisements influence people on spending decisions by
(a) Using special music (b) Using attractive pictures
(c) Making them think that they need the item (d) All the above
Ans: (d) All the above

16. When I go shopping, I will buy
(a) Something that looks attractive (b) Something my friend has
(c) Something that I need to purchase (d) The first thing I see in the store
Ans: (c) Something that I need to purchase

17. The best shopping choice is to
- (a) Shop at brand name stores always buy
 - (b) Compare the choices before buying
 - (c) The same thing my friends bought
 - (d) Buy at a regular shop always
- Ans: (b) Compare the choices before buying

18. Praveen recently got the registration number for his new two wheeler. Here the number is given in the form of mirror image. Encode the image and find the correct registration number

T N 1 2 H 2 5 8 9

- (a) **6 8 5 2 H 2 1 N 1** (b) **1 N 1 5 H 5 2 8 9**
(c) **9 8 5 2 H 5 1 N 1** (d) **9 8 5 2 H 2 1 N T**

Ans: Option (a)

19. Which of the following stage of cryptography are the readable non-encrypted data?

(a) Plain Text (b) Encryption (c) Cipher Text (d) Decryption

Ans: (a) Plain Text

20. Which of the following stage of cryptography are the non-readable encrypted data?

(a) Plain Text (b) Encryption (c) Cipher Text (d) Both (b)&(c)

Ans: (c) Cipher Text

21. In how many ways does information in cryptography is encrypted?

(a) 2 (b) 3 (c) 4 (d) 5

Ans: (a) 2

22. Which of the following term defines the string of char that assist during encryption and decryption of confidential message?

(a) Cipher (b) Plain text (c) Key (d) All the above

Ans: (c) Key

23. Which of the following are the components of crypto system?

(a) Plain text (b) Cipher text (c) Keys (d) All the above

Ans: (d) All the above

24. What is the first term (F1) and second term (F2) in the standard Fibonacci sequence?

(a) 0, 1 (b) 1, 1 (c) 1, 2 (d) 0, 0

Ans: (a) 0, 1

Some variations start 1, 1, but 0, 1 is common

25. In the sequence 0, 1, 1, 2, 3, 5, ____, which number comes next?

(a) 6 (b) 7 (c) 8 (d) 9

Ans: (c) 8

26. Which of these numbers is NOT in the Fibonacci sequence (0, 1, 1, 2, 3, 5, 8, 13...)?

- (a) 13 (b) 21 (c) 15 (d) 34

Ans: (c) 15

27. The general formula for the Fibonacci sequence is:

- (a) $F(n) = F(n-1) - F(n-2)$ (b) $F(n) = F(n-1) + F(n-2)$
(c) $F(n) = F(n-2) \times F(n-1)$ (d) $F(n) = F(n-1) + 1$

Ans: (b) $F(n) = F(n-1) + F(n-2)$

28. If the sequence is 3, 5, 8, 13, ____, what is the next term?

- (a) 18 (b) 21 (c) 26 (d) 34

Ans: (b) 21

Use the Atbash Cipher code to find the correct words for the below questions (29 - 33)

Plain Text	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher Text	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

29. To find the correct word **G Z N R O**

- (a) MATIL (b) TAMIL (c) MILTA (d) TILMA

Ans: (b) TAMIL

30. To find the correct word **V M T O R H S**

- (a) ENGLISH (b) GLIESHN (c) NGHLIES (d) LISHENG

Ans: (a) ENGLISH

31. To find the correct word **N Z G S V N Z G R X H**

- (a) HATIMATCSMA (b) MATMAHAICST (c) MATHAMATICS (d) None

Ans: (c) MATHAMATICS

32. To find the correct word **H L X R Z O H X R V M X V**

- (a) SOCIAL SCIENCE (b) SCIECNE SOCILA
(c) SOCIAL SCIENCE (d) None of these

Ans: (c) SOCIAL SCIENCE

33. To find the correct word **H X R V M X V**

- (a) ENSCICE (b) SCIENCE (c) ENCICSE (d) None

Ans: (b) SCIENCE

Use the below Reflection Table to find the correct words for the below questions (34 - 37)

A	B	C	D	E	F	G	H	I	J	K	L	M
↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

34. J V A Q B J

- (a) WODNIW (b) WODWIN (c) WINDOW (d) None of these

Ans: (c) WINDOW

35. P B Z C H G R E G N O Y R

- (a) ELBATRET UPMOC (b) TAEBCOP TUERT
(c) COMPUTER TABLE (d) None of these

Ans: (c) COMPUTER TABLE

36. P U N V E

- (a) AIRHC (b) CHAIR (c) RIAHC (d) None of these

Ans: (b) CHAIR

37. P H O O B N E Q

- (a) CUBBOARD (b) DRAOBBUC (c) BBORDAUC (d) None of these

Ans: (a) CUBBOARD

38. The word "Doog Gninrom" is given in the form of mirror image. Encode the image and find the correct word.

- (a) Good gninrom (b) Good Morning (c) Morning good (d) None

Ans: (b) Good Morning

Use the below Pigpen Cipher text to find the correct words for the below questions (39 - 41)

1A	2B	3C	10J	11K	12L	19S	23W
4D	5E	6F	13M	14N	15O	20T	24X
7G	8H	9I	16P	17Q	18R	21U	25Y
						22V	26Z

39. The room number in which the treasure took place: $\sqcup \sqcap$

- (a) 26 (b) 82 (c) 48 (d) 28

Ans: (d) 28

40. Place of the treasure : $\sqcup \sqcap \sqcup \sqcap \sqcap$

- (a) JQARI (b) CHAIR (c) AHRIQ (d) None of these

Ans: (b) CHAIR

41. The name of the treasure : 

- (a) TJFG RECHUOV (b) GFTJ CHUREOV
(c) GIFT VOUCHER (d) None of these

Ans: (c) GIFT VOUCHER

Use the below Pigpen Cipher text to find the correct words for the below questions (42 - 44)

A	B	C	J	K	L	S	W
D	E	F	M	N	O	T	X
G	H	I	P	Q	R	U	Y
						V	Z

42. Decode the treasure using Pigpen Cipher text



- (a) RETAW ELTTOB (b) WATER BOTTLE
(c) TERAW ELTBOT (d) None of these

Ans: (b) WATER BOTTLE

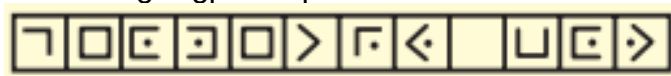
43. Decode the treasure using Pigpen Cipher text



- (a) MAXE DAP (b) AXEM ADP (c) EXAM PAD (d) None of these

Ans: (c) EXAM PAD

44. Decode the treasure using Pigpen Cipher text



- (a) GEOMETRY BOX (b) YRTEM OEG XOB
(c) METGEORY OBX (d) None of these

Ans: (a) GEOMETRY BOX

45. Suppose the first Fibonacci number is 0 and the second is 1. What is the sixth Fibonacci number?

- (a) 5 (b) 6 (c) 7 (d) 8

Ans: (a) 5

46. Which of the following is not a Fibonacci number?

- (a) 8 (b) 21 (c) 55 (d) 14

Ans: (d) 14

47. When a die is thrown, what is the probability of getting a number greater than 5?

- (a) $1/8$ (b) $1/3$ (c) $1/6$ (d) $3/2$

Ans: (c) $1/6$

48. What is the probability of getting product of number greater than 25 when 2 dice are rolled?

- (a) $11/36$ (b) $1/36$ (c) $1/12$ (d) $3/2$

Ans: (c) $1/12$

Total Outcomes $6 \times 6 = 36$

Favorable Outcomes (5, 6) (6, 5) (6, 6)

$$P(\text{product} > 25) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} = \frac{3}{36} = \frac{1}{12}$$

49. A bag has 5 red, 3 white and 2 black balls. What is the probability of picking a black or red ball from bag?

- (a) $2/10$ (b) $7/10$ (c) $1/4$ (d) $2/15$

Ans: (b) $7/10$

50. What is the probability of getting non-prime number when a die is rolled?

- (a) $1/2$ (b) $1/3$ (c) $1/4$ (d) $1/6$

Ans: (a) $1/2$

51. What is the probability of getting all heads when 4 coins are tossed?

- (a) $1/2$ (b) $1/4$ (c) $1/8$ (d) $1/16$

Ans: (d) $1/16$

52. A bag has 23 balls; 12 red, 5 blue and 6 green. What is the probability of picking a green ball from bag?

- (a) $12/23$ (b) $6/23$ (c) $5/23$ (d) $1/23$

Ans: (b) $6/23$

53. A card is picked from well shuffled deck of 52 cards. What is the probability of picking a club card?

- (a) $1/2$ (b) $1/13$ (c) $1/4$ (d) $1/26$

Ans: (c) $1/4$

$$P(\text{Club}) = 13/52 = 1/4.$$

54. What is the probability of drawing a card of value greater than 5 and lesser than 9 from deck of cards?

- (a) $1/52$ (b) $12/52$ (c) $13/52$ (d) $1/26$

Ans: (b) $12/52$

55. What is the probability of getting 2 tails and 1 head when 3 coins are tossed?

- (a) $1/2$ (b) $1/13$ (c) $3/8$ (d) $1/3$

Ans: (c) 3/8

56. What is the probability of picking an ace of hearts card from deck of cards?

- (a) $1/52$ (b) $1/13$ (c) $1/4$ (d) $13/52$

Ans: (a) 1/52

57. Choose the correct card to complete the perfect SET of all different on each card.



Ans:



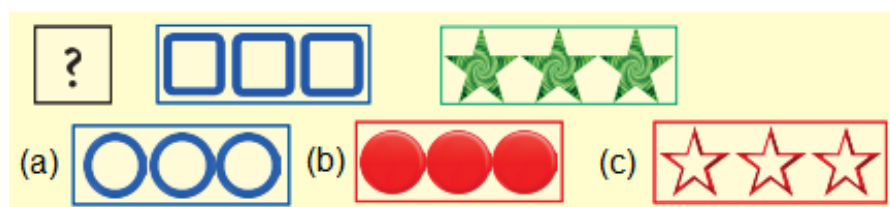
58. Choose the correct card to complete the perfect SET of all different on each card.



Ans:



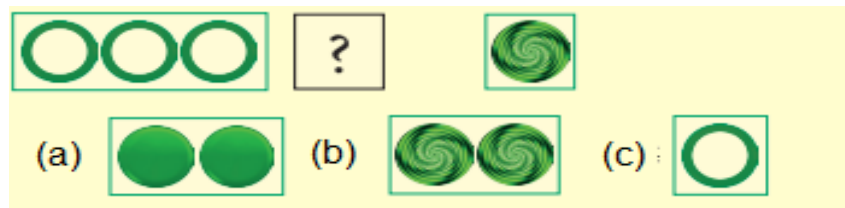
59. Choose the correct card to complete the perfect SET of all different on each card.



Ans:



60. Choose the correct card to complete the perfect SET of all different on each card.



Ans: (a) 

61. Choose the correct card to complete the perfect SET of all different on each card.



Ans: (b) 

62. What are the prime factors of 144?

- (a) 12×12 (b) $2 \times 3 \times 2 \times 3 \times 2 \times 2$ (c) $3 \times 2 \times 3 \times 2 \times 3 \times 2$ (d) $2^4 \times 3^3$

Ans: (b) $2 \times 3 \times 2 \times 3 \times 2 \times 2$

63. Which of these is the prime factorization of 42?

- (a) $2 \times 3 \times 7$ (b) 3×14 (c) 2×21 (d) 6×7

Ans: (a) $2 \times 3 \times 7$

64. What number should be multiplied by 392 to make it a perfect cube?

- (a) 2 (b) 7 (c) 9 (d) 4

Ans: (b) 7

65. Which number is NOT prime?

- (a) 17 (b) 23 (c) 29 (d) 33

Ans: (d) 33

66. What are the common prime factors of 256 and 156?

- (a) 2, 3 (b) 2, 13 (c) 2 (d) 1

Ans: (c) 2

67. What are the prime factors of 144?

- (a) 12×12 (b) $12 \times 2 \times 6$ (c) $2 \times 3 \times 2 \times 3 \times 2 \times 2$ (d) $3 \times 2 \times 3 \times 2 \times 3 \times 2$

Ans: (c) $2 \times 3 \times 2 \times 3 \times 2 \times 2$

68. Find the prime factors of 1000.

- (a) $2 \times 5 \times 2 \times 5 \times 5 \times 5$ (b) $2 \times 5 \times 2 \times 5 \times 2 \times 5$ (c) $2 \times 2 \times 5 \times 2 \times 5 \times 2$ (d) $4 \times 5 \times 5 \times 2 \times 5$

Ans: (b) $2 \times 5 \times 2 \times 5 \times 2 \times 5$

69. Which of the prime factors would be correct for square of 9?

- (a) $3 \times 3 \times 9$ (b) 3^4 (c) 9×9 (d) 8×1

Ans: (b) 3^4

70. Find the square root of 225 by using prime factors.

- (a) 3×5 (b) 5×5 (c) 3×3 (d) 15

Ans: (a) 3×5

71. Prime factors of 12 are _____

- (a) 3×4 (b) 3×2 (c) $3 \times 2 \times 2$ (d) $4 \times 2 \times 4$

Ans: (c) $3 \times 2 \times 2$

72. Two different numbers can have same prime factors.

- (a) True (b) False (c) Can't determine (d) None of these

Ans: (b) False

73. Square root of 324 is _____

- (a) 18 (b) 12 (c) 22 (d) 24

Ans: (a) 18

74. Prime factors of 41 are _____

- (a) 1×41 (b) 4×1 (c) $1 \times 4 \times 2$ (d) $2 \times 2 \times 1$

Ans: (a) 1×41

75. What are the prime factors of prime numbers?

- (a) The prime number itself.
(b) Any number depending on the prime number
(c) Prime number and the number itself
(d) There cannot be any prime factors

Ans: (c) Prime number and the number itself

76. What are prime numbers?

- (a) A number which has one and only one factor
(b) A with only 1 as factor
(c) A number which has 1 and the number itself as the factors
(d) All numbers are prime numbers

Ans: (c) A number which has 1 and the number itself as the factors

77. A prime number

- (a) Has one factor
(b) Has more than 2 factors
(c) Has 100 factors
(d) Has exactly 2 factors: 1 and the number

Ans: (d) Has exactly 2 factors: 1 and the number

78. What is the HCF of two consecutive odd numbers?

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (a) 1

79. What is the HCF of two consecutive even numbers?

- (a) 1 (b) 2 (c) 3 (d) 4

Ans: (b) 2

80. $2 \times 2 \times 2$ or 2^3 is the prime factorization of

- (a) 6 (b) 12 (c) 22 (d) 8

Ans: (d) 8

81. Which number below is NOT PRIME?

- (a) 2 (b) 11 (c) 18 (d) 13

Ans: (c) 18

82. Which is a PRIME number?

- (a) 15 (b) 17 (c) 21 (d) 39

Ans: (b) 17

83. Which is a COMMON FACTOR between 12 and 18?

- (a) 4 (b) 3 (c) 9 (d) 5

Ans: (b) 3

84. Write 144 in prime factorization. Show in a factor tree.

- (a) $2^2 \times 3^4$ (b) $2^4 \times 3^2$ (c) $2^3 \times 3^3$ (d) $2^4 \times 3^3$

Ans: (b) $2^4 \times 3^2$

85. Find the prime factorization of 72. Show in a factor tree.

- (a) $2^3 \times 3^2$ (b) $2 \times 3^2 \times 4$ (c) $2^4 \times 3^3$ (d) $2^2 \times 6^2$

Ans: (b) $2 \times 3^2 \times 4$

86. Prime factorisation of 54 is

- (a) 2×27 (b) $2 \times 3 \times 9$ (c) 54×1 (d) $2 \times 3 \times 3 \times 3$

Ans: (d) $2 \times 3 \times 3 \times 3$

87. $1 \times 2 \times 3 \times 7$ is the prime factorisation of

- (a) 21 (b) 1237 (c) 237 (d) 42

Ans: (d) 42

88. The LCM and HCF of two numbers are equal. The numbers must be:

- (a) Co-prime (b) Prime (c) Equal (d) Both even

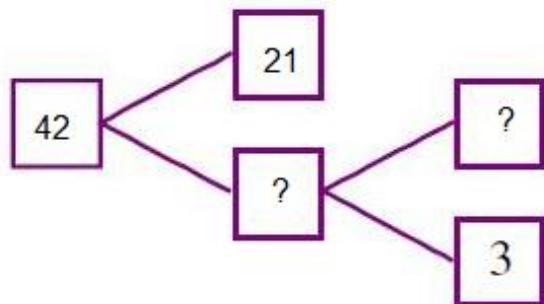
Ans: (c) Equal

89. If x and y are co-primes, then their LCM is

- (a) 1 (b) $x \times y$ (c) x / y (d) $1 / 2 \times y$

Ans: (b) $x \times y$

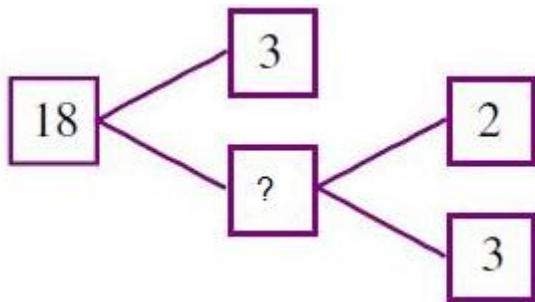
90. In the given factor tree, what are the missing numbers?



- (a) 2, 7 (b) 6, 7 (c) 2, 3 (d) 4, 2

Ans: (a) 2, 7

91. The missing number in the following factor tree is



- (a) 7 (b) 5 (c) 6 (d) 4

Ans: (c) 6

92. Find the common prime factor of 256 and 156.

- (a) 2 (b) 3 (c) 4 (d) 1

Ans: (a) 2

$256 = 2^8$, $156 = 2^2 \times 3 \times 13$. The only common prime factor is 2

93. What is the smallest number to multiply with 392 to get a perfect cube?

- (a) 7 (b) 2 (c) 9 (d) 4

Ans: (a) 7

94. The HCF of two prime numbers is always:

- (a) 0 (b) 1 (c) The numbers themselves (d) Their product

Ans: (b) 1

95. The LCM of 8 and 12 is:

- (a) 4 (b) 12 (c) 24 (d) 48

Ans: (c) 24

96. The LCM of two numbers is 2079 and their HCF is 27. If one number is 189, the other is:

- (a) 297 (b) 584 (c) 189 (d) 216

Ans: (a) 297

$$2079 \times 27 = 189 \times \text{Other} \Rightarrow \text{Other} = 297$$

97. Two bells ring at intervals of 12 and 18 minutes. When will they ring together next after starting together?

- (a) 24 min (b) 30 min (c) 36 min (d) 72 min

Ans: (c) 36 min

98. Find the LCM of 9, 12, and 15 using prime factorization:

- (a) 150 (b) 120 (c) 90 (d) 180

Ans: (d) 180

$$\begin{aligned} \text{Prime factors: } 9 &= 3^2, 12 = 2^2 \times 3, 15 = 3 \times 5 \\ \text{LCM} &= 2^2 \times 3^2 \times 5 = 4 \times 9 \times 5 = 180. \end{aligned}$$

99. The LCM of 36 and 144 is

- (a) 36 (b) 144 (c) 16 (d) 18

Ans: (b) 144

100. What is the LCM of two consecutive odd numbers?

- (a) Addition of those two numbers
(b) Subtraction of those two numbers
(c) Product of those two numbers
(d) Division of those two numbers

Ans: (c) Product of those two numbers